



CONTACT

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SKILLS

- Instrumentation**
Clamp-on ultrasonic flowmeters · Meter verification · Correction factors
- Embedded**
STM32 · ESP32 · Arduino · RTOS · I²C / UART / PWM
- Software**
Python · C / C++ · C# · JavaScript / React · MATLAB
- AI & Signals**
PyTorch · TensorFlow · OpenCV · DSP · EEG / EMG · PID control
- Security**
Burp Suite · Nmap · Wireshark · Metasploit
- Tools & CAD**
KiCad · SolidWorks · Fusion 360 · Git · Linux · Docker

LANGUAGES

- Spanish** Native
- English** Professional · C1

INTERESTS

- Formula 1 · Road trips · The ocean · Gaming · Photography

Andy Casafranca.

Mechatronics Engineer · Instrumentation, Embedded Systems & AI

PROFILE

Mechatronics Engineering graduate (B.Sc., UPC, Lima), currently working as an Instrumentation Engineer on field **flow measurement** for Ica's water utility. Also designs **embedded systems**, trains **AI models** and delivers end-to-end IoT solutions, with bilingual experience supporting U.S. clients.

EXPERIENCE

Sep 2026 — Present Ica, Perú · On-site	Instrumentation Engineer LabProjects · Temporary contract - Validate and control flow measurement across the wells and reservoirs of EMAPICA, Ica's water utility. - Benchmark installed meters against clamp-on ultrasonic flowmeters and adjust their correction factors so field data stays accurate.
Feb — Jun 2025 Remote	Bilingual Customer Support Agent Major U.S. last-mile delivery platform - Resolved order, routing and payment issues for delivery drivers in real time over chat and phone, in English and Spanish. - Met strict response-time and quality targets at high volume and de-escalated time-sensitive incidents.
2023 — 2024 Lima, Perú	Undergraduate Research Assistant Universidad Peruana de Ciencias Aplicadas (UPC) - Designed a custom analog front-end and STM32 firmware for an EEG/EMG acquisition headset. - Built a Python (NumPy, SciPy) signal pipeline reaching 87% accuracy on motor-imagery classification across 8 channels.
Feb — Aug 2023 Lima, Perú · On-site	Bilingual Customer Support Agent Major U.S. health-insurance network - Handled claims, eligibility and benefits inquiries from U.S. healthcare providers under strict regulatory protocols. - Delivered accurate bilingual support in a high-volume, regulated call-center environment.
2022 — 2023 Remote · Freelance	Embedded Systems & Automation Engineer Independent projects - Delivered three turnkey IoT projects (hardware, firmware and dashboards), including a smart-greenhouse mesh and a water-quality monitor. - Wrote ESP32 mesh firmware in C/C++ with OTA updates, paired with React dashboards over WebSocket. - Ran a responsible-disclosure security assessment of a consumer IoT camera, resulting in a CVE write-up with a proof-of-concept exploit.
SELECTED PROJECTS	
In progress	HydraDetect-AI Clandestine water-tap detection Detects illegal water connections in city pipes using a controlled water hammer and machine learning: 89% field accuracy on an offline Arduino edge device.
2017	Hackatón MINSA · 1st place Ministry of Health · Team Vincere Co-built a game-based app for preliminary depression screening with the 10-student team that won Peru's national health hackathon.
EDUCATION	
2019 — 2026 Graduated	B.Sc. in Mechatronics Engineering Universidad Peruana de Ciencias Aplicadas (UPC) · Lima, Perú Focus: control systems, embedded firmware, biomedical instrumentation and applied AI.
2024 Online	Practical Ethical Hacking & Network Security Self-directed · TCM Security · TryHackMe Hands-on penetration-testing labs covering the OWASP Top 10 and Active Directory exploitation paths.